DEFINITION

Medication error can be defined as 'a failure in the treatment process that leads to, or has the potential to lead to, harm to the patient'.

These are not adverse drug reactions

Medication errors can occur in:

- choosing a medicine—irrational, inappropriate, and ineffective prescribing, underprescribing and overprescribing;
- writing the prescription—prescription errors, including illegibility;
- manufacturing the formulation to be used—wrong strength, contaminants or adulterants, wrong or misleading packaging;
- dispensing the formulation—wrong drug, wrong formulation, wrong label;
- administering or taking the drug—wrong dose, wrong route, wrong frequency, wrong duration;
- monitoring therapy—failing to alter therapy when required, erroneous alteration.

11.9 Systems of Medication Measurement

Metric system: Most commonly used and convenient system. Basic units of measurements are metre, litre and gram.

Apothecary system: Infrequently used and basic unit of measurements are grain, minim.

Household system: Least accurate and used only in houses. Basic units of measurements are teaspoon, tablespoon.

Conversion within systems:

1g = 1000mg, 1L = 1000ml

Metric	Apothecary	Household
1ml	15 to 16 minim	15 drops
4-5ml	1 fluidram	1 tsp
15-16ml	4 fluidram	1 tbsp
30ml	1 ounce	2 tbsp
240ml	8 ounce	1 cup
480 ml (apprx. 500 ml)	1 pint	1 pint
960 ml (apprx. 1 lt.)	1 quart	1 quart
4800 ml (apprx. 5 lt.)	1 gallon	1 gallon

11.10 Routes of Medication Administration

There are 5 major routes of medication administration that includes:

I. Oral route:

- Oral
- Enteral (through enteral tube)
- Buccal (placing between cheek and gum)
- Sublingual (placing under tongue)

II. Parenteral route:

- Intradermal (under epidermis)

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